

Converification

Non-Convex Problem:

$$v(\boldsymbol{\theta}) = \min\{f(\mathbf{x}; \boldsymbol{\theta}) : \mathbf{x} \in \mathcal{C}(\boldsymbol{\theta})\}$$

Convex Relaxation:

$$v_r(\boldsymbol{\theta}) \leq v(\boldsymbol{\theta})$$

Deployment

Fast Non-Convex Solver:

Feasible solution $\hat{\mathbf{x}} \in \mathcal{C}(\boldsymbol{\theta})$
with no optimality guarantee

If $f(\hat{\mathbf{x}}; \boldsymbol{\theta}) \leq \hat{v}(\boldsymbol{\theta}) - \hat{q}_\alpha + \delta$,
certify $\hat{\mathbf{x}}$ as δ -optimal with
probability at least $1 - \alpha$

Training
 $\mathcal{E}$
Calibration

Deep Neural Network:

$$\hat{v}(\boldsymbol{\theta}) \approx v_r(\boldsymbol{\theta}),$$
$$\mathbb{P}(v_r(\boldsymbol{\theta}) \geq \hat{v}(\boldsymbol{\theta}) - \hat{q}_\alpha) \geq 1 - \alpha$$